

GlucoTwin

A personalised Digital Twin for Type 2 Diabetes that predicts glucose spikes 2 hours before they happen

Team Mavericks · NIT Silchar · Digital Twin Challenge 2026, Happiest Health

Diabetes care in India is reactive

101M

people with diabetes in India (ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023)

Most of them have Type 2 Diabetes

A check-up every 3 months

HbA1c shows the average, not the daily spikes after meals

Indian lifestyle drives spikes

Rice/roti-heavy meals, late dinners, short sleep, low activity

Data overload

A CGM produces 288 readings a day — no doctor can read them for 500 patients

Our idea: a living virtual copy of each patient



Static stream

EHR: age, BMI, HbA1c, labs, meds, TCF7L2 gene



Dynamic stream

Every 5 min: CGM glucose, heart rate, HRV, steps, sleep, meal log



Digital Twin

Fuses both streams + personal physiology model



Doctor gets

2-h spike risk, forecast, reasons, alerts, what-if

Proactive, not reactive: act before glucose crosses 180 mg/dL

Data: synthetic, privacy-safe, India-specific

Synthetic EHR (FHIR-style)

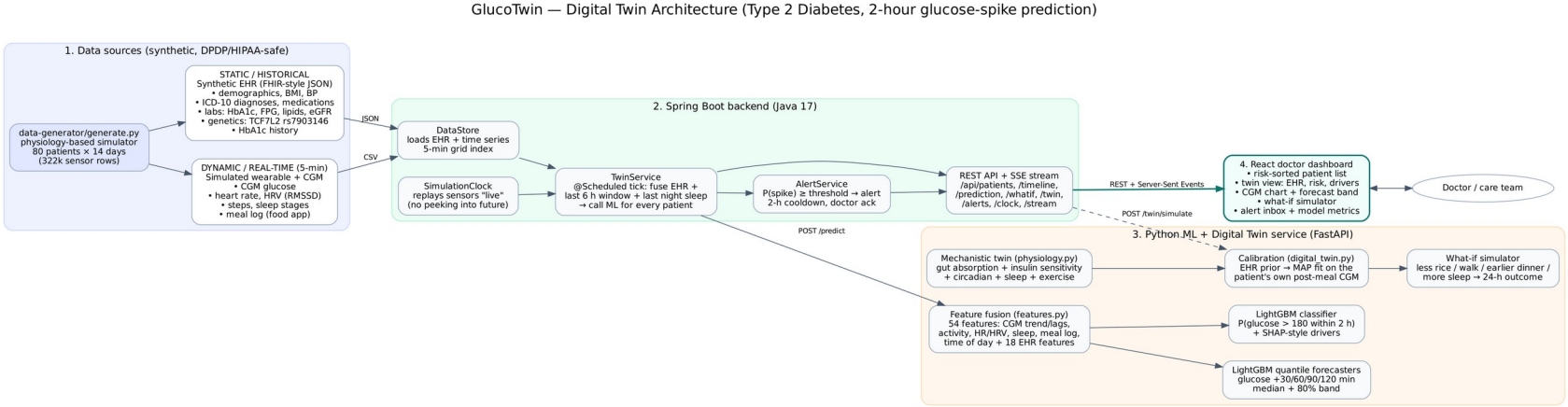
- 80 virtual Indian adults with T2D
- ICD-10 diagnoses: E11, I10, E78.5
- Metformin, glimepiride, basal insulin, statins
- HbA1c, FPG, PPBG, lipids, eGFR, urine ACR
- Genetics: TCF7L2 rs7903146, family history

Simulated wearable + CGM

- 14 days × 288 readings/day = 322k rows
- CGM with 10-min lag, noise, dropouts
- Heart rate, HRV, steps, sleep stages
- Meal log: ~70% of meals, ±22% carb error
- Late dinners, rice-heavy lunches, festive sweets

Labs are consistent with each patient's simulated glucose (HbA1c from the ADAG formula). No real patient data is used (DPDP Act / HIPAA). The schema maps to Synthea / MIMIC-IV.

Architecture



Python data generator | **FastAPI + LightGBM twin** | **Spring Boot 3 (Java 17)** | **React + Recharts**

Fusion model: 54 features, 3 jobs

17 — **CGM**
level, lags, trend, variability

16 — **Wearable + app**
steps, HR, HRV, sleep, meals

18 — **EHR**
HbA1c, BMI, meds, genes

3 — **Time**
time of day, weekend



Spike classifier

LightGBM: P(glucose > 180 within 2 h). Threshold set for ≥ 85% sensitivity on validation patients



Glucose forecaster

12 quantile models: +30/60/90/120 min, median + 80% band



Explanations

TreeSHAP contributions grouped into plain words: "carbs logged", "short sleep"

The twin: small, explainable physiology

```
carbs → gut (2 stages) → blood

dG/dt = cf·Ra − k·S·(G − Gb) − exercise + stress

S changes with: evening dip · sleep debt · walking

Gb rises with the dawn phenomenon (4–8 am)
```

1. **EHR prior**
   FPG → basal glucose; BMI, gene, HbA1c → insulin sensitivity
2. **Calibrate**
   Fit to the patient's own post-meal CGM curves (5 days)
3. **Simulate**
   Run the next 24 h under a doctor's what-if scenario

0.52 → 0.63

insulin sensitivity: correlation with hidden truth

0.84 → 0.88

basal glucose: correlation with hidden truth

67%

patients with a better fit on 2 held-out days

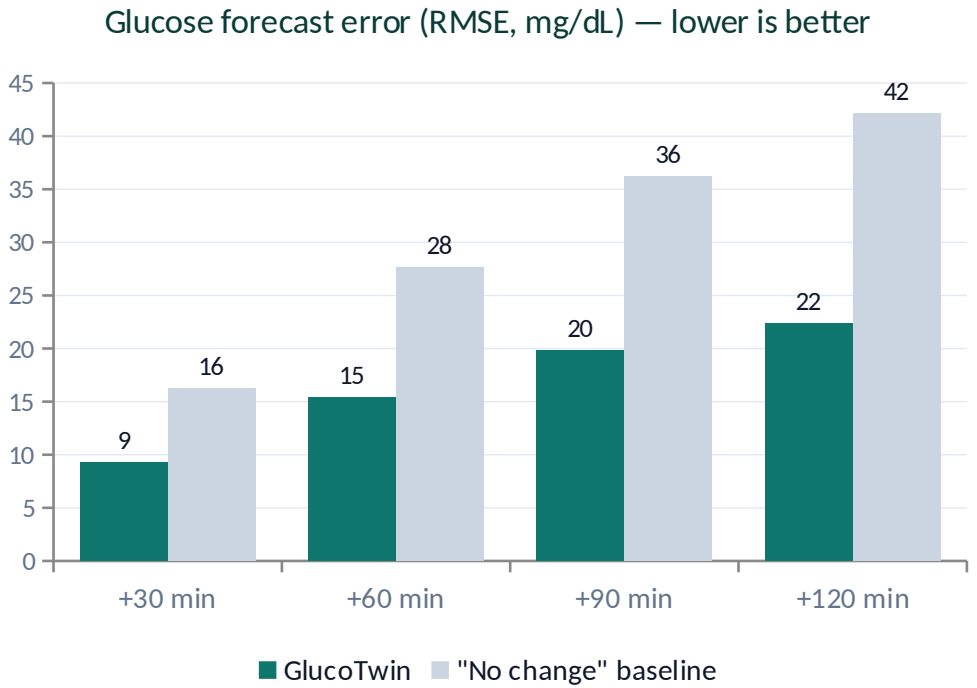
Results on 24 patients the model never saw

0.95 AUROC

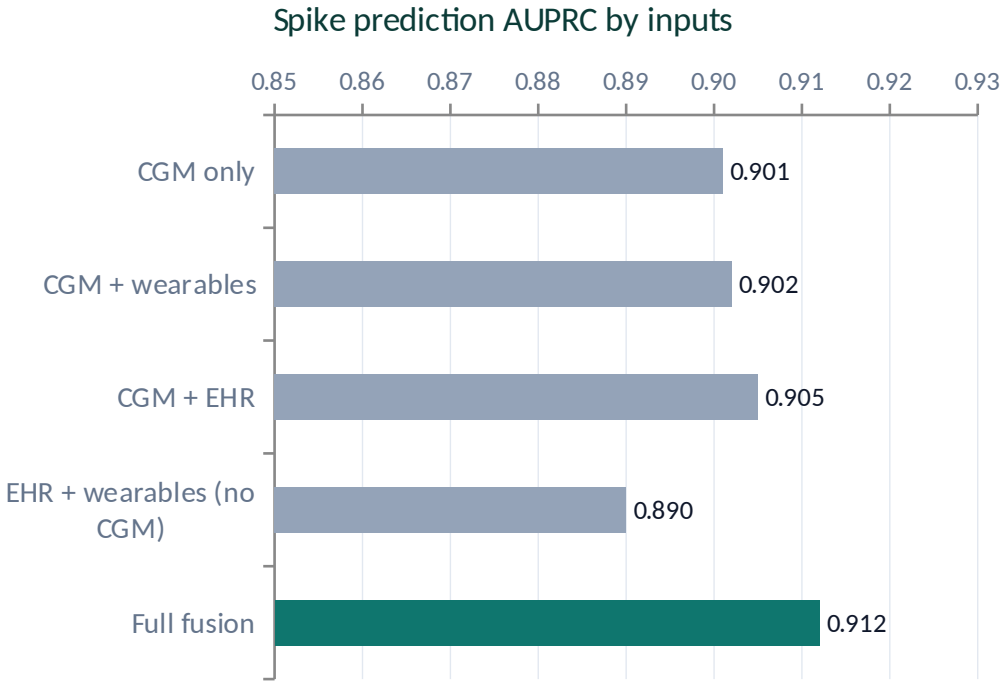
85.6% spikes flagged ≥ 30 min early

100 min median warning time

2.15 false alarms per patient-day



Does fusing the two streams help?



Full fusion is best

0.912 vs 0.901 for CGM alone

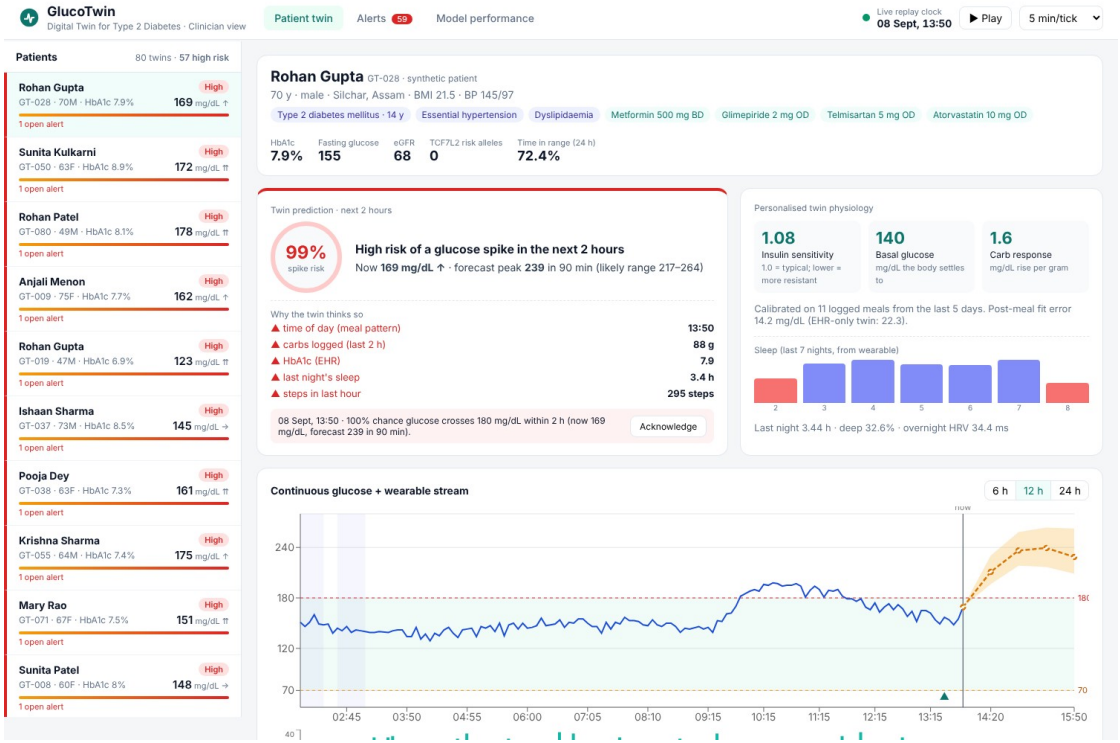
Works without a CGM

EHR + a basic smartwatch still reach 0.890, useful where CGMs are too costly

Honest note

Gains are small here: short-term glucose is driven mostly by recent CGM values and meal timing

The doctor's view



Risk-sorted list

Highest-risk patients first, with live glucose and trend arrow

2-hour prediction

Probability, forecast peak, and why

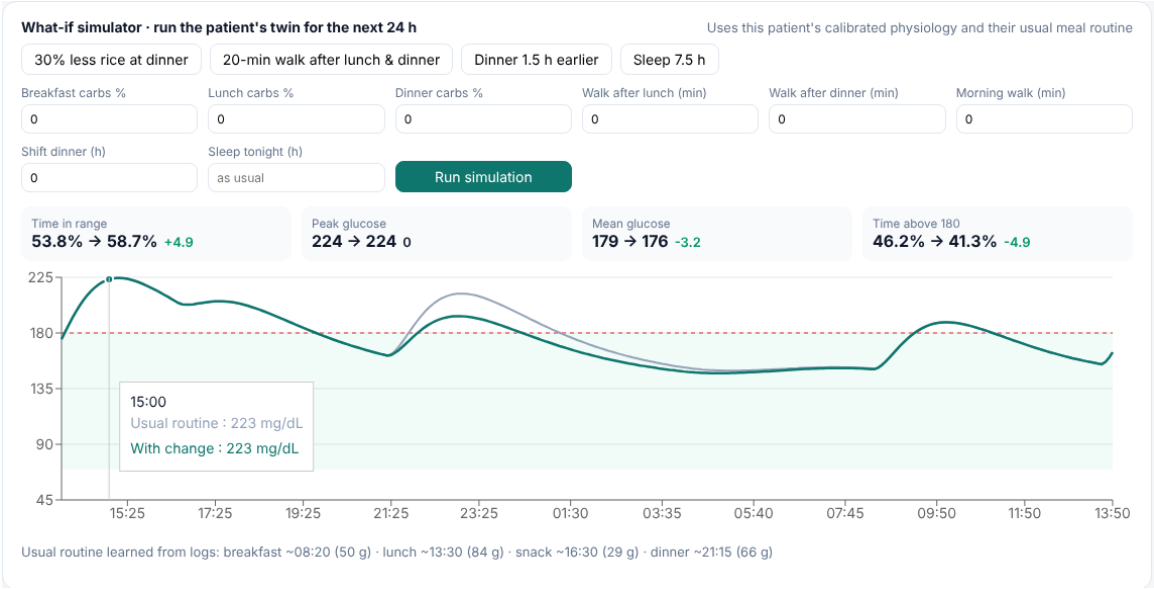
Live stream chart

CGM, forecast band, meals, steps, sleep

Alert inbox

Acknowledge with an action note

What-if: test advice on the twin first



Example: 30% less rice at dinner

+4.9

percentage points of time-in-range for this patient (53.8% → 58.7%) over the next 24 h

Other levers: post-meal walks, earlier dinner, more sleep, morning walk

Real-world impact and fit for India



Remote monitoring at scale

One nurse can watch hundreds of twins and act only on alerts



Prevent, don't treat

A 15-minute walk advised 90 minutes early can blunt a spike



Low-cost path

Works with a smartwatch + EHR when a CGM isn't available



Privacy by design

Synthetic sandbox now; ABDM/FHIR + DPDP-consent integration next

Limitations and roadmap

Limitations (honest)

- Synthetic data from the same model family, so the accuracy numbers are optimistic
- Unlogged meals limit calibration
- Not a medical device; needs clinical validation

1 **Validate on real data**

CGMacros (PhysioNet): T2D CGM + Fitbit + meals

2 **Meal detection**

Infer unlogged meals from CGM rises

3 **Integrate**

ABDM / FHIR EHR pull with consent

4 **Close the loop**

Patient nudges + measure spikes prevented



Thank you

Team Mavericks · NIT Silchar

Raushan Kumar (team leader) · Chaitanya Prasad · EIE, final year

Code under the MIT License · synthetic data only